

ST3101: 3. Exercises on all of linear algebra

Note that some of the [Slide references \(in blue\)](#) may change as slide decks are updated

1. Matrix addition and multiplication

(a) Let $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ be the first, second, and third column of

$$\mathbf{X} = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 1 & 5 \\ 3 & 2 & 7 \end{bmatrix}$$

respectively. Find

$$\mathbf{x}_1 - \mathbf{x}_2$$

(b) For $\mathbf{X}$ as in 1(a), find

$$\mathbf{x}_1 + \mathbf{x}_2$$

(c) Assume that the columns of $\mathbf{X}$ as in 1(a) are observations on three variables. A researcher wants you for each observation $i = 1, 2, 3$, to create a new variable

$$y_i = 3x_{i1} + x_{i2} - 2x_{i3}$$

Write this in the $\mathbf{Ax} = \mathbf{b}$ form (with all the numbers in the matrices)

(d) Find v in

$$\begin{bmatrix} \frac{1}{3} & -\frac{1}{2} \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} -4/3 & -2/3 \\ 8 & v \end{bmatrix}$$

(e) Find v in

$$\begin{bmatrix} 1 & 1 \\ -1 & \frac{1}{2} \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 3 \\ 0 & 0 \\ 2 & v \end{bmatrix}$$

(f) Find v in

$$\begin{bmatrix} 2 & 1 & 0 \\ 3 & 1 & 6 \\ 7 & 1 & 9 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 5 & 2 \\ 0 & 6 & 2 \\ 1 & 7 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 16 & 6 \\ 9 & 63 & 20 \\ 16 & 104 & 34 \\ v & 45 & 14 \end{bmatrix}$$

(g) Find $\mathbf{C}$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 1 \\ 5 & 6 & 7 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 & 7 & 1 \\ 1 & 1 & 1 & 2 \\ 0 & 6 & 9 & 4 \end{bmatrix}$$

(h) Find $\mathbf{C}$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 1 \\ 5 & 6 & 7 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 1 \\ 0 & 6 \end{bmatrix}$$

(i) For

$$\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 3 & 6 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 3 & -2 & 1 \\ 2 & 4 & 5 \end{bmatrix}$$

(1) Find $\mathbf{AB}$

(2) Find $\mathbf{BA}$

(a)

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$$

(c)

$$\mathbf{y} = \mathbf{Xb}$$

for

$$\mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$$

check

$$\mathbf{Xb} = 3 \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 1 \times \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} - 2 \times \begin{bmatrix} 5 \\ 5 \\ 7 \end{bmatrix}$$

(d) These are all systems

$$\mathbf{AB} = \mathbf{C}$$

To find c_{22} , we note that the second column of $\mathbf{C}$ is given by

$$\mathbf{Ab}_2 = 1 \times \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} + 2 \times \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -2/3 \\ v \end{bmatrix}$$

and hence we only need to look at the bottommost equation

$$2 + 2 = v$$

(e) To find c_{32} , we note that the second column of $\mathbf{C}$ is given by

$$\mathbf{A}\mathbf{b}_2 = 1 \times \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} + 2 \times \begin{bmatrix} 1 \\ \frac{1}{2} \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ v \end{bmatrix}$$

and hence we only need to look at the bottommost equation

$$3 - 2 = v$$

(f) To find c_{41} , we note that the first column of $\mathbf{C}$ is given by

$$\mathbf{A}\mathbf{b}_1 = 1 \times \begin{bmatrix} 2 \\ 3 \\ 7 \\ 1 \end{bmatrix} + 0 \times \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix} + 1 \times \begin{bmatrix} 0 \\ 6 \\ 9 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 9 \\ 16 \\ v \end{bmatrix}$$

and hence we only need to look at the bottommost equation

$$1 + 4 = v$$

2. Trace

See Lecture 3 matrices slides [43-46](#) for explanations and definition.

(a) For $\mathbf{X}$ as in 1(a), find

$$\text{tr}(\mathbf{X})$$

(b) For $\mathbf{X}$ as in 1(a), find

$$\text{tr}(\mathbf{X}^\top)$$

(c) For $\mathbf{X}$ as in 1(a), and

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 5 & 6 & 7 \\ 2 & 2 & 2 \end{bmatrix}$$

let $\mathbf{C} = \mathbf{A} + \mathbf{X}$. Find

$$\text{tr}(\mathbf{C})$$

(d) For $\mathbf{A}$ and $\mathbf{B}$ as in 1(i), let $\mathbf{C} = 2\mathbf{A}\mathbf{B}$. Find

$$\text{tr}(\mathbf{C})$$

(e) For $\mathbf{A}$ and $\mathbf{B}$ as in 1(i), let $\mathbf{C} = 2\mathbf{A}\mathbf{B}$. Find

$$\text{tr}(\mathbf{C}^\top)$$

(a)

$$\text{tr}(\mathbf{X}) = \sum_{i=1}^3 x_{ii} = 1 + 1 + 7 = 9$$

(b) We could (but do not have to) define $\mathbf{A} = \mathbf{X}^\top$, which, by definition has $a_{ij} = x_{ji}$ and

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^3 a_{ii} = \sum_{i=1}^3 x_{ii} = 1 + 1 + 7 = 9$$

(c) With

$$\begin{bmatrix} 1 & 0 & 1 \\ 5 & 6 & 7 \\ 2 & 2 & 2 \end{bmatrix}$$

the trace is $\text{tr}(\mathbf{A}) = 1 + 6 + 2 = 9$. The matrix $\mathbf{C} = \mathbf{A} + \mathbf{X}$ is given by

$$\mathbf{A} + \mathbf{X} = \begin{bmatrix} 1 & 0 & 1 \\ 5 & 6 & 7 \\ 2 & 2 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 3 & 5 \\ 2 & 1 & 5 \\ 3 & 2 & 7 \end{bmatrix} = \begin{bmatrix} 1+1 & 0+3 & 1+5 \\ 5+2 & 6+1 & 7+5 \\ 2+3 & 2+2 & 2+7 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 6 \\ 7 & 7 & 12 \\ 5 & 4 & 9 \end{bmatrix}$$

and the trace is

$$\text{tr}(\mathbf{C}) = 2 + 7 + 9 = 18$$

but

$$\text{tr}(\mathbf{A}) + \text{tr}(\mathbf{X}) = 9 + 9 = 18$$

as it should be. In fact we know that

$$\text{tr}(\mathbf{C}) = \sum_{i=1}^3 c_{ii} = \sum_{i=1}^3 (a_{ii} + x_{ii}) = \sum_{i=1}^3 a_{ii} + \sum_{i=1}^3 x_{ii} = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{X})$$

(d) Letting $\mathbf{a}_i$ be the i 'th row of $\mathbf{A}$, and $\mathbf{b}_j$ be the j 'th column of $\mathbf{B}$, from the definition of matrix products, we know that we can write element in row i and column j of $\mathbf{AB}$ as

$$\mathbf{a}_i \mathbf{b}_j = \sum_{k=1}^3 a_{ik} b_{kj}$$

Applying the definition of the trace we have

$$\text{tr}(\mathbf{AB}) = \sum_{i=1}^3 \mathbf{a}_i \mathbf{b}_i = \sum_{i=1}^3 \sum_{k=1}^2 a_{ik} b_{ki}$$

here

$$\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 3 & 6 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 3 & -2 & 1 \\ 2 & 4 & 5 \end{bmatrix}$$

hence

$$\text{tr}(\mathbf{AB}) = \begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} -2 \\ 4 \end{bmatrix} + \begin{bmatrix} 3 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \end{bmatrix} = 9 - 8 + 33 = 34$$

And $\text{tr}(\mathbf{C}) = 68$

(e)

$$\mathbf{C}^\top = 2(\mathbf{AB})^\top = 2\mathbf{B}^\top \mathbf{A}^\top$$

and the trace

$$\text{tr}(\mathbf{AB}^\top) = \sum_{i=1}^3 \mathbf{b}_i^\top \mathbf{a}_i = \sum_{i=1}^3 \sum_{k=1}^2 b_{ki} a_{ik}$$

which we see is identical to $\text{tr}(\mathbf{AB})$. But to be sure here

$$\mathbf{A}^\top = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -1 & 6 \end{bmatrix} \text{ and } \mathbf{B}^\top = \begin{bmatrix} 3 & 2 \\ -2 & 4 \\ 1 & 5 \end{bmatrix}$$

hence

$$\text{tr}(\mathbf{AB}^\top) = \begin{bmatrix} 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} -2 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \end{bmatrix} = 9 - 8 + 33 = 34$$

And $\text{tr}(\mathbf{C}^\top) = 68$

3. Data manipulation

This exercise is extensive and problems 4 and 5 are more important at this stage of the course. In this course we will not do much data manipulation manually but here are some exercises on how we can manipulate data matrices using linear

algebra. We will return to the matrix of sum of squares and cross-products later when we do linear discriminant analysis.

(a) For $\mathbf{X}$ as in 1(a), find

$$(\mathbf{x}_1 - \mathbf{x}_2)^\top (\mathbf{x}_1 - \mathbf{x}_2)$$

(b) For $\mathbf{X}$ as in 1(a), find

$$\sum_{i=1}^3 x_{i1}^2$$

(c) For $\mathbf{X}$ as in 1(a), find

$$\sum_{i=1}^3 (x_{i1} - x_{i2})^2$$

(d) Let $\bar{\mathbf{X}} = \frac{1}{n} \mathbf{X}^\top \mathbf{1}_{n \times 1}$, and demonstrate that

$$\mathbf{1}_{3 \times 1} \bar{\mathbf{X}}^\top = \frac{1}{n} \mathbf{1}_{3 \times 1} \mathbf{1}_{3 \times 1}^\top \mathbf{X}$$

i.e. show numerically that the left-hand side is equal to the right-hand side as in

$$\mathbf{X} - \mathbf{1}_{n \times 1} \bar{\mathbf{X}}^\top = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} - \begin{bmatrix} \bar{X}_1 & \bar{X}_2 & \cdots & \bar{X}_p \\ \bar{X}_1 & \bar{X}_2 & \cdots & \bar{X}_p \\ \vdots & \vdots & \ddots & \vdots \\ \bar{X}_1 & \bar{X}_2 & \cdots & \bar{X}_p \end{bmatrix}$$

(e) Using $n = 2$ and

$$\mathbf{1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

show numerically that the expression

$$\frac{1}{n^2} \mathbf{1} \mathbf{1}^\top \mathbf{1} \mathbf{1}^\top = \frac{1}{n} \mathbf{1} \mathbf{1}^\top$$

is true.

(f) Using the rule that $\mathbf{ABC} = \mathbf{A}(\mathbf{BC})$, show that the expression

$$\frac{1}{n^2} \mathbf{1} \mathbf{1}^\top \mathbf{1} \mathbf{1}^\top = \frac{1}{n} \mathbf{1} \mathbf{1}^\top$$

is true for general n .

(g) Assume $p = 2$ and that the following covariance matrix is provided

$$\mathbf{S} = \begin{bmatrix} 9.33 & 17.33 \\ 17.33 & 37.33 \end{bmatrix}$$

Define $\mathbf{S} = \frac{1}{n-1} \mathbf{SSCP}$, where the sum of squares and cross-products is

$$\mathbf{SSCP} = \mathbf{X}^\top \left(\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^\top \right) \mathbf{X} = \begin{bmatrix} \sum_{i=1}^n (x_{i1} - \bar{X}_1)^2 & \sum_{i=1}^n (x_{i1} - \bar{X}_1)(x_{i2} - \bar{X}_2) & \cdots & \sum_{i=1}^n (x_{i1} - \bar{X}_1)(x_{ip} - \bar{X}_p) \\ \sum_{i=1}^n (x_{i1} - \bar{X}_1)(x_{i2} - \bar{X}_2) & \sum_{i=1}^n (x_{i2} - \bar{X}_2)^2 & \cdots & \sum_{i=1}^n (x_{i2} - \bar{X}_2)(x_{ip} - \bar{X}_p) \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^n (x_{i1} - \bar{X}_1)(x_{ip} - \bar{X}_p) & \sum_{i=1}^n (x_{i2} - \bar{X}_2)(x_{ip} - \bar{X}_p) & \cdots & \sum_{i=1}^n (x_{ip} - \bar{X}_p)^2 \end{bmatrix}$$

Find the correlation matrix from $\mathbf{S}$, using that

$$\mathbf{R} = \frac{1}{n-1} \mathbf{SSCP}_s = \mathbf{D}^{-1/2} \mathbf{SD}^{-1/2}$$

where

$$\mathbf{SSCP}_s = \mathbf{D}^{-1/2} \underbrace{\mathbf{X}^\top \left(\mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) \mathbf{X}}_{\text{crossprod. centered}} \mathbf{D}^{-1/2}$$

and

$$\mathbf{D}^{1/2} = \begin{bmatrix} s_1 & 0 & \cdots & 0 \\ 0 & s_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & s_p \end{bmatrix}$$

is a matrix with the standard deviations on the diagonal.

(a) Using what we have calculated above

$$(\mathbf{x}_1 - \mathbf{x}_2)^\top (\mathbf{x}_1 - \mathbf{x}_2) = \begin{bmatrix} -2 & 1 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = 4 + 1 + 1 = 6$$

We could also use algebra to simplify by expanding the expression

$$(\mathbf{x}_1 - \mathbf{x}_2)^\top (\mathbf{x}_1 - \mathbf{x}_2) = \mathbf{x}_1^\top \mathbf{x}_1 - 2\mathbf{x}_2^\top \mathbf{x}_1 + \mathbf{x}_2^\top \mathbf{x}_2$$

where

$$\mathbf{x}_1^{\text{top}} \mathbf{x}_1 = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 + 4 + 9 = 14$$

and

$$\mathbf{x}_2^\top \mathbf{x}_2 = \begin{bmatrix} 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} = 9 + 1 + 4 = 14$$

and

$$\mathbf{x}_1^\top \mathbf{x}_2 = \begin{bmatrix} 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 3 + 2 + 6 = 11$$

so that

$$(\mathbf{x}_1 - \mathbf{x}_2)^\top (\mathbf{x}_1 - \mathbf{x}_2) = 14 - 22 + 14 = 6$$

(b) We know from one of our first exercises, we know that we have the definition of inner product as

$$\mathbf{x}_1^\top \mathbf{x}_1 = \sum_{i=1}^3 x_{1i}^2$$

and in the previous solution we calculated $\mathbf{x}_1^\top \mathbf{x}_1 = 14$

(c) We know from one of our first exercises, we know that we have the definition of inner product for any vector

$$\mathbf{a}^\top \mathbf{a} = \sum_{i=1}^3 a_i^2$$

and in the previous solution we calculated $\mathbf{a}^\top \mathbf{a}$ for $\mathbf{a} = \mathbf{x}_1 - \mathbf{x}_2$

$$(\mathbf{x}_1 - \mathbf{x}_2)^\top (\mathbf{x}_1 - \mathbf{x}_2) = \begin{bmatrix} -2 & 1 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = 4 + 1 + 1 = 6$$

(d) Data

$$\mathbf{X} = \begin{bmatrix} 2 & 8 \\ 4 & 10 \\ 6 & 12 \end{bmatrix}$$

and mean vector

$$\bar{\mathbf{X}} = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$$

The matrix

$$\mathbf{1}_{3 \times 1} \bar{\mathbf{X}}^\top$$

must have 3 rows and two columns, with the first column being

$$4 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

and the second

$$10 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

putting it together

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} [4 \quad 10] = \begin{bmatrix} 4 & 10 \\ 4 & 10 \\ 4 & 10 \end{bmatrix}$$

For the RHS, first

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} [1 \quad 1 \quad 1] = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

multiplying this by $\mathbf{X}$

$$\mathbf{1}_{3 \times 1} \mathbf{1}_{3 \times 1}^\top \mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 8 \\ 4 & 10 \\ 6 & 12 \end{bmatrix}$$

yields a 3×2 matrix, whose first column is

$$2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 4 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 6 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+4+6 \\ 2+4+6 \\ 2+4+6 \end{bmatrix} = \begin{bmatrix} 12 \\ 12 \\ 12 \end{bmatrix}$$

similarly for the second column, which will have $8 + 10 + 12 = 30$ in each row, so that

$$\frac{1}{n} \mathbf{1}_{3 \times 1} \mathbf{1}_{3 \times 1}^\top \mathbf{X} = \frac{1}{3} \begin{bmatrix} 12 & 30 \\ 12 & 30 \\ 12 & 30 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 4 & 10 \\ 4 & 10 \end{bmatrix}$$

(e) We start with $\mathbf{1}\mathbf{1}^\top$ in the RHS

$$\mathbf{1}\mathbf{1}^\top = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

On the LHS the product

$$\mathbf{1}\mathbf{1}^\top \mathbf{1}\mathbf{1}^\top = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1^2 + 1^2 & 1^2 + 1^2 \\ 1^2 + 1^2 & 1^2 + 1^2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

Using that $n = 2$

$$\frac{1}{4} \mathbf{1}\mathbf{1}^\top \mathbf{1}\mathbf{1}^\top = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2} \mathbf{1}\mathbf{1}^\top$$

(f) Looking at the LHS, we see that we can write

$$\mathbf{1}\mathbf{1}^\top \mathbf{1}\mathbf{1}^\top = \mathbf{1}(\mathbf{1}^\top \mathbf{1})\mathbf{1}^\top$$

and the inner product

$$\mathbf{1}^\top \mathbf{1} = \sum_{i=1}^n 1 \times 1 = n$$

consequently

$$\frac{1}{n^2} \mathbf{1}\mathbf{1}^\top \mathbf{1}\mathbf{1}^\top = \frac{1}{n^2} \mathbf{1}n\mathbf{1}^\top = \frac{n}{n^2} \mathbf{1}\mathbf{1}^\top = \frac{1}{n} \mathbf{1}\mathbf{1}^\top$$

(g) This rests on the fact that if we post-multiply a matrix with a diagonal matrix, we will multiply each column with a constant, e.g.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1/s & 0 \\ 0 & 1/t \end{bmatrix} = \begin{bmatrix} a/s & b/t \\ c/s & d/t \end{bmatrix}$$

If we pre multiply a matrix with a diagonal matrix

$$\begin{bmatrix} 1/s & 0 \\ 0 & 1/t \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a/s & b/s \\ c/t & d/t \end{bmatrix}$$

we will multiply each row with its diagonal element. Combining these, we have

$$\begin{bmatrix} 1/s & 0 \\ 0 & 1/t \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1/s & 0 \\ 0 & 1/t \end{bmatrix} = \begin{bmatrix} a/(ss) & b/(st) \\ c/(st) & d/(tt) \end{bmatrix}$$

so that each element a_{ij} is divided by $d_i d_j$. To get the correlation

$$\text{corr}(X_i, X_j) = \frac{\text{cov}(X_i, X_j)}{\sqrt{V(X_i)}\sqrt{V(X_j)}}$$

we want to divide each element s_{ij} of $\mathbf{S}$ with the product of the diagonal elements $\sqrt{s_{ii}}$ and $\sqrt{s_{jj}}$. Hence

$$\begin{aligned} \begin{bmatrix} \frac{1}{3.05} & 0 \\ 0 & \frac{1}{6.11} \end{bmatrix} \begin{bmatrix} 9.33 & 17.33 \\ 17.33 & 37.33 \end{bmatrix} \begin{bmatrix} \frac{1}{3.05} & 0 \\ 0 & \frac{1}{6.11} \end{bmatrix} &= \begin{bmatrix} \frac{9.33}{3.05} & \frac{17.33}{3.05 \times 6.11} \\ \frac{17.33}{3.05 \times 6.11} & \frac{37.33}{6.11} \end{bmatrix} \\ &= \begin{bmatrix} 1 & .93 \\ .93 & 1 \end{bmatrix} \end{aligned}$$

4. The determinant

(a) Using the calculation formula (slide 36 Lecture 3 Matrices) for the determinant of 2×2 matrices, show that

$$\begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab$$

(b) Find

$$\begin{vmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2} \end{vmatrix}$$

(c) Using Property 1 (slide 28 Lecture 3 Matrices) and Property 3 a (slide 30 Lecture 3 Matrices), show that

$$\begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} = abc$$

(d) Find

$$\begin{vmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

(e) Find

$$\begin{vmatrix} 3 & 2 \\ 0 & 1 \end{vmatrix}$$

(f) Find

$$\begin{vmatrix} 3 & 0 \\ 2 & 1 \end{vmatrix}$$

(g) Find

$$\begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix}$$

(h) Find

$$\begin{vmatrix} 3 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

(i) Find

$$\begin{vmatrix} 3 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

(j) Find

$$\begin{vmatrix} 3 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix}$$

(k) Find

$$\begin{vmatrix} 3 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

(a) Here you do not have to use ‘the formula’ because we know from **Properties** 1 and 3(a) that the determinant of a diagonal matrix is the product of diagonal elements. Nevertheless, using the formula

$$\begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab - 0 \times 0 = ab$$

(b) Here we will only use that the determinant of a diagonal matrix is the product of diagonal elements

$$\begin{vmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2} \end{vmatrix} = \sqrt{2}\sqrt{2} = 2$$

(c) Successively applying **Property 3**, we have

$$\begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} = a \begin{vmatrix} 1 & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} = ab \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & c \end{vmatrix} = abc \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

and applying property 1, we have that the determinant is abc .

(d)

$$\begin{vmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 3 \times 1 \times 1 = 3$$

(e) Using **Property 5**, we have

$$\begin{vmatrix} 3 & 2 \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} 3 - 2 \times 0 & 2 - 2 \times 1 \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} 3 & 0 \\ 0 & 1 \end{vmatrix} = 3 \times 1 = 3$$

as, previously we established that the determinant of a diagonal matrix is the product of the diagonal elements. **IN FACT**, you can now prove that:

The determinant of a upper triangular matrix is the product of the diagonal elements

How do we know this? Well, we know that we can use elimination to create a diagonal matrix from a triangular, using only subtraction of rows. Consider and $n \times n$ matrix, the sub matrix in the lower right corner will have elements

$$\begin{bmatrix} a_{(n-1)(n-1)} & a_{(n-1)n} \\ a_{n(n-1)} & a_{nn} \end{bmatrix} = \begin{bmatrix} a_{(n-1)(n-1)} & a_{(n-1)n} \\ 0 & a_{nn} \end{bmatrix}$$

Subtracting

$$\frac{a_{(n-1)n}}{a_{nn}}$$

times row n from row $n - 1$ will result in the submatrix being

$$\begin{bmatrix} a_{(n-1)(n-1)} & a_{(n-1)n} - \frac{a_{(n-1)n}}{a_{nn}}a_{nn} \\ 0 & a_{nn} \end{bmatrix} = \begin{bmatrix} a_{(n-1)(n-1)} & 0 \\ 0 & a_{nn} \end{bmatrix}$$

By property 5, this does not change the determinant. Having done this for row $n - 1$, we can proceed to do it for every other row, until all rows, above row n , have zeros in their last column. Then we can do it for element $n - 1$ in row $n - 1$, and so on.

(f) We know from property 10 that $\det(\mathbf{A}^\top) = \det(\mathbf{A})$. Using the solution to (e):

$$\begin{vmatrix} 3 & 0 \\ 2 & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 0 & 1 \end{vmatrix} = 3$$

We therefore have a straightforward corollary to the determinant of and upper triangular matrix: the determinant of a triangular matrix is the product of diagonal elements.

(g) Apply the rule of the transpose first

$$\begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix}$$

and then apply Property 5

$$\begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} = \begin{vmatrix} 3 - 2 \times 1 & 2 - 2 \times 1 \\ 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix}$$

This is clearly a triangular matrix again so

$$\begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1$$

(h) Again, this is a triangular matrix so the determinant is $3 \times 1 \times 1 = 3$. Let us try applying property 5 again though:

$$\begin{vmatrix} 3 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \{L1 - L3 \mapsto L1\} = \begin{vmatrix} 3 - 0 & 1 - 0 & 1 - 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 3 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Now, subtract row 2 from row 1

$$\begin{vmatrix} 3 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \{L1 - L2 \mapsto L1\} = \begin{vmatrix} 3 - 0 & 1 - 1 & 0 - 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

and we have ourselves a diagonal matrix with determinant 3.

(i) Again, this is a triangular matrix so the determinant is $3 \times 2 \times 1 = 6$. Let us try applying property 5 again though:

$$\begin{vmatrix} 3 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \{L1 - L3 \mapsto L1\} = \begin{vmatrix} 3-0 & 1-0 & 1-1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 3 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Now, subtract half of row 2 from row 1

$$\{L1 - \frac{1}{2}L2 \mapsto L1\} = \begin{vmatrix} 3 - \frac{1}{2} \times 0 & 1 - \frac{1}{2} \times 2 & 0 - \frac{1}{2} \times 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

and we have ourselves a diagonal matrix with determinant 6.

(j) We want to turn this into a triangular matrix. We start by getting rid of the 1 in row 3 column 1

$$\begin{vmatrix} 3 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = \{L3 - \frac{1}{3}L1 \mapsto L3\} = - \begin{vmatrix} 3 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & -\frac{1}{3} & \frac{2}{3} \end{vmatrix}$$

Now, add a third of row 2 to row 3

$$\{L3 + \frac{1}{3}L2 \mapsto L3\} = \begin{vmatrix} 3 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{2}{3} \end{vmatrix} = 3 \times 1 \times \frac{2}{3} = 2$$

and we have ourselves a diagonal matrix with determinant 2. Can you find an easier route?

(k) This matrix has a row of zeros so we know that it cannot be invertible and thus should have determinant 0. It is also a triangular matrix with determinant $3 \times 0 \times 1 = 0$.

5. The matrix inverse

(a) Find the inverse of $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix}$$

(b) With $\mathbf{A}$ as in 5(a) let $\mathbf{B} = \mathbf{A}^{-1}$. What is $\mathbf{B}^{-1}$?

(c) Find the inverse of $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

(d) Find the inverse of $\mathbf{U}$

$$\mathbf{U} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

using (1) elimination as on [slide 22-23](#) of Lecture 3 Matrices, and (2) using the cofactor method on [slide 41](#) of Lecture 3 Matrices.

The formula for 3×3 (without the typo) is:

For 3×3 matrix $\mathbf{A}$

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj}(\mathbf{A}^\top)$$

first take the transpose of $\mathbf{A}$

$$\mathbf{A}^\top = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

then the **adjoint** (elements are cofactors)

$$\text{adj}(\mathbf{A}^\top) = \begin{bmatrix} + \begin{vmatrix} a_{22} & a_{32} \\ a_{23} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{12} & a_{32} \\ a_{13} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{12} & a_{22} \\ a_{13} & a_{23} \end{vmatrix} \\ - \begin{vmatrix} a_{21} & a_{31} \\ a_{23} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{31} \\ a_{13} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{21} \\ a_{13} & a_{23} \end{vmatrix} \\ + \begin{vmatrix} a_{21} & a_{31} \\ a_{22} & a_{32} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{31} \\ a_{12} & a_{32} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{vmatrix} \end{bmatrix}$$

(e) Determine if the inverse of $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

exist.

(f) Determine if the inverse of $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

exist.

(f) Find the inverse of $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}.$$

(a) We know that the inverse of a diagonal matrix is a diagonal matrix with the reciprocals on the diagonal, so

$$\mathbf{A}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix}^{-1} = \mathbf{A}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/6 \end{bmatrix}$$

(b)

$$\mathbf{B}^{-1} = (\mathbf{A}^{-1})^{-1} = \mathbf{A}$$

Check using the actual numbers, i.e., check that

$$\begin{bmatrix} 1 & 0 \\ 0 & 1/6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(c) We can do this using elimination as well as using the formula. First elimination

operations augmented matrix

$$\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{array}$$

$$L_1 - 2L_2 \rightarrow L_1 \quad \begin{array}{cc|cc} 1 & 0 & 1 & -2 \\ 0 & 1 & 0 & 1 \end{array}$$

and we are done and

$$\mathbf{A}^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

We can check this, which is easy as the first column will just be the first column of the first matrix

$$\begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \times 1 + 1 \times (-2) \\ 0 & 2 \times 0 + 1 \times 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Using the formula, we first note that the determinant is 1, and thus the inverse is simply

$$\mathbf{A}^{-1} = \text{adj}(\mathbf{A}^\top) = \begin{bmatrix} a_{11} & -a_{12} \\ -a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

(d) To calculate $\mathbf{U}^{-1}$, we see that we can easily get rid of the top-most 1 in $\mathbf{U}$

operations augmented matrix

$$\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}$$

$$L_1 - L_3 \rightarrow L_1 \quad \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}$$

Hence,

$$\mathbf{U}^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

In general for triangular matrices, it is easy to get the inverse. The principle is the same as for calculating the determinant for a triangular matrix.

(e) This matrix has a last row of 0 so $\mathbf{A}$ is not invertible and has determinant of 0. We could use the rule for triangular matrices and see that the product of diagonal elements are 0.

(f) If we do elimination, we try to first get rid of the 1st and second elements of the third row and then see where we go from there

operations	augmented matrix					
	1	2	3	1	0	0
	0	1	4	0	1	0
	5	6	0	0	0	1
$L_3 - 5L_1 \rightarrow L_2$	1	2	3	1	0	0
	0	1	4	0	1	0
	0	-4	-15	-5	0	1
$L_3 + 4L_2 \rightarrow L_2$	1	2	3	1	0	0
	0	1	4	0	1	0
	0	0	1	-5	4	1
$L_1 - 3L_3 \rightarrow L_1$	1	2	0	16	-12	-3
	0	1	4	0	1	0
	0	0	1	-5	4	1
$L_2 - 4L_3 \rightarrow L_2$	1	2	0	16	-12	-3
	0	1	0	20	-15	-4
	0	0	1	-5	4	1
$L_1 - 2L_2 \rightarrow L_1$	1	0	0	-24	18	5
	0	1	0	20	-15	-4
	0	0	1	-5	4	1

We can check in R that this is correct. This would not have been a course in linear algebra (which it isn't) if you had not been forced to also apply the formula

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj}(\mathbf{A}^\top)$$

Do it and see if you get the same answer.